

Rogowski Coils Suit Relay Protection and Measurement

Ljubomir Kojovic*

Microprocessor-based technology has been accepted today in virtually all areas of electrical engineering, including power. Input signals that accurately replicate events are critical for the proper operation of microprocessor-based equipment. Therefore, current sensors are an integral element of relay protection and current measuring systems.

Current transformers (CTs) have been used exclusively for protection and measurement applications in part because of their ability to produce the high power output needed by electromechanical equipment. The introduction of microprocessor-based equipment has made the high power output of CTs unnecessary and made the use of other measurement techniques possible.

One such measuring device that does not produce high power output but offers many advantages over CTs is the Rogowski coil. This article describes Rogowski coil operational characteristics and its applications for current measurements.

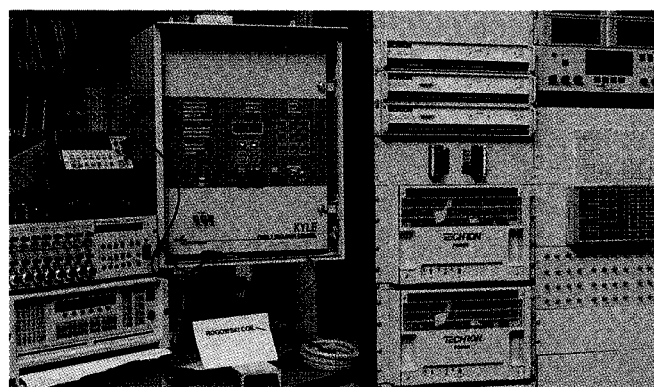
Measuring Coils vs Current Transformers

Current transformers have been widely used for relay protection and current measurement. For protection applications, CT saturation is a major concern. When CTs saturate, secondary signals become distorted and may cause relay misoperation. The dc component of the primary current is the dominant factor causing the buildup of flux in the CT core. The flux created by the dc component is usually much larger than the flux needed to drive the ac component alone. Therefore, faults with dc components (asymmetrical currents) are more likely to saturate CTs. CTs can also retain an unknown amount of flux. This remanent flux will either oppose or aid the buildup of the core flux, depending on the relative polarities of the primary current dc component and the remanent flux. Gapless CTs can reproduce both ac and dc components in the CT secondary proportional to the primary current up to the CT knee-point voltage. CTs with antiremanence gaps are less susceptible to saturation. They can reproduce both the ac and dc components in the CT secondary proportional to the instantaneous value of the primary current up to design accuracy limiting conditions. Air-gapped CTs can be used in applications in

which the accurate reproduction of the dc component in the secondary circuit is of no interest. Remanent flux is substantially reduced in these CTs. Measurements and computer simulations have shown that CT errors are small in the frequency range of 1 Hz to several thousand Hz. An important characteristic of CTs is their high output power (1 A or 5 A secondary current and tens or hundreds VA) which enables them to drive electro-mechanical equipment.

Today, it is evident, that there is an irreversible trend toward using microprocessor-based equipment for relay protection and measurements. The equipment does not

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require high power current and voltage sensors. Even though their high powered outputs are no longer necessary, because CTs and PTs have been used for many years, microprocessor relays have been designed to accept signals from these conventional instrument transformers. This is done by using scaling transformers at the equipment inputs, which transform standard secondary currents and voltages to the required low voltage levels. If the input transformers are removed and the equipment redesigned for the direct application of the voltages needed for their operation, Rogowski coils could be used as current sensors instead of CTs.

*Cooper Power Systems

Rogowski coils have many advantages over conventional CTs that include:

- High measurement accuracy: They can be designed to be better than 0.1 percent, but are typically 1-3 percent.
- Wide measurement range: The same coil can be used to measure currents from several amps to hundreds of kiloamps. Rogowski coils do not measure direct currents, but, unlike CTs, they can accurately measure currents when a large dc component is present

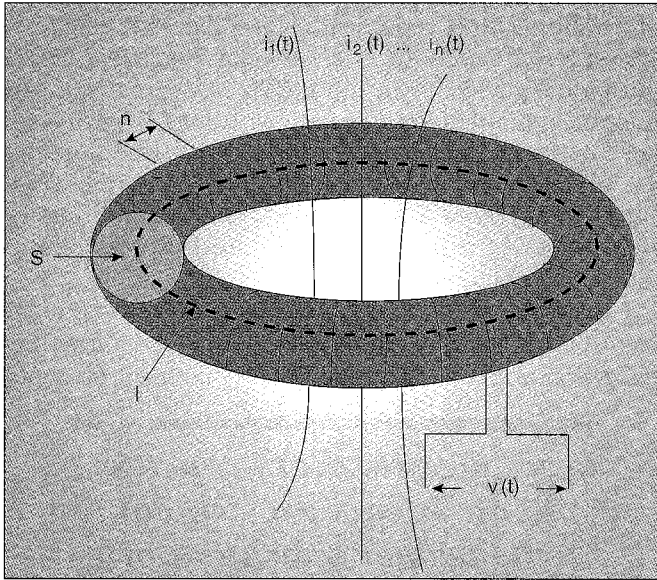


Figure 1. Principle of Rogowski coil design

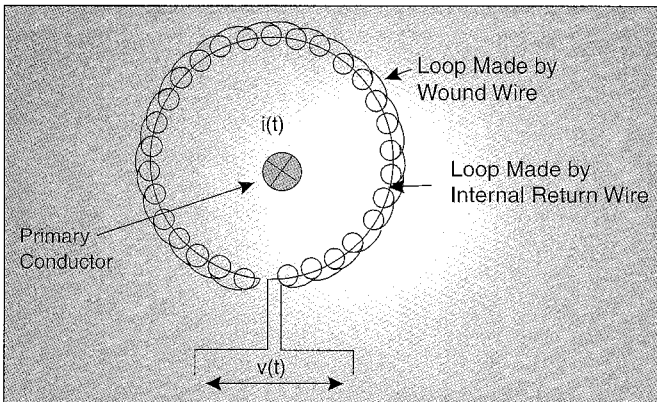


Figure 2. Rogowski coil with the return wire through the winding center (single-arm design)

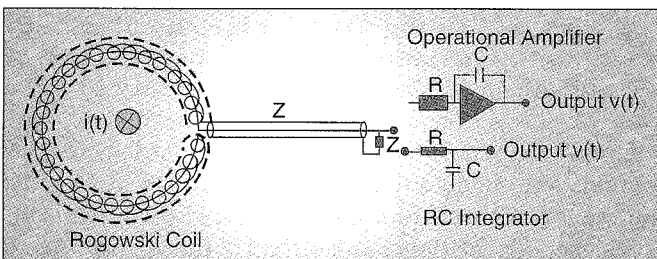


Figure 3. Rogowski coil output voltage integration using an RC integrator or an operational amplifier

because there is no iron core to saturate.

- Wide frequency range: Typically 0.1 Hz to over 1 MHz (depending on design). Specially designed coils can have a bandpass frequency of up to 200 MHz.
- Can withstand unlimited short-circuit currents.
- Can be very small to allow measurement of currents in restricted areas where other techniques cannot be used.
- Can be flexible or rigid depending on application requirements.
- Can be used to measure current distributions in circuits having very small impedances without affecting the circuits.
- Galvanically isolated from the primary conductor.
- Can drive digital protection relays and measuring devices such as voltmeters, oscilloscopes, transient recorders, and data loggers.
- Have a low production cost.

Theory of Operation

Rogowski coils consist of wire wound on a nonmagnetic core ($\mu=\mu_0$). The coil is placed around the conductors whose currents are to be measured as shown in Figure 1.

The voltage induced in the coil is defined by Equation 1.

$$v(t) = -\frac{d}{dt} \left(\sum_{j=1}^N \Phi_j \right) \quad (1)$$

where Φ_j is instantaneous flux for the j -th turn of the total N turns. The total flux in the coil is given in Equation 2.

$$\Psi = \oint_l \int_s \vec{H} n d\vec{S} \quad (2)$$

where $v(t)$ is the coil output voltage, ψ is the total flux, l is the core length, n is the winding density (windings per unit length), S is the core cross-section, and H is the magnetic field intensity.

If the core has a constant cross-section and the wire is wound perpendicular on the middle line l with constant density, the coil output voltage is then:

$$v(t) = -\mu_0 n S \frac{d}{dt} \left[\sum_j i_j(t) \right] = -M \frac{d}{dt} \left[\sum_j i_j(t) \right] \quad (3)$$

where μ_0 is the permeability of free space, $i(t)$ is the primary current, and M is the coil mutual reactance.

Rogowski coil output voltage is proportional to the rate of change of the measured current. If the design conditions stated previously are followed, the coil mutual reactance M is independent of the conductor location inside the coil loop. To prevent the influence of nearby conductors carrying high currents, Rogowski

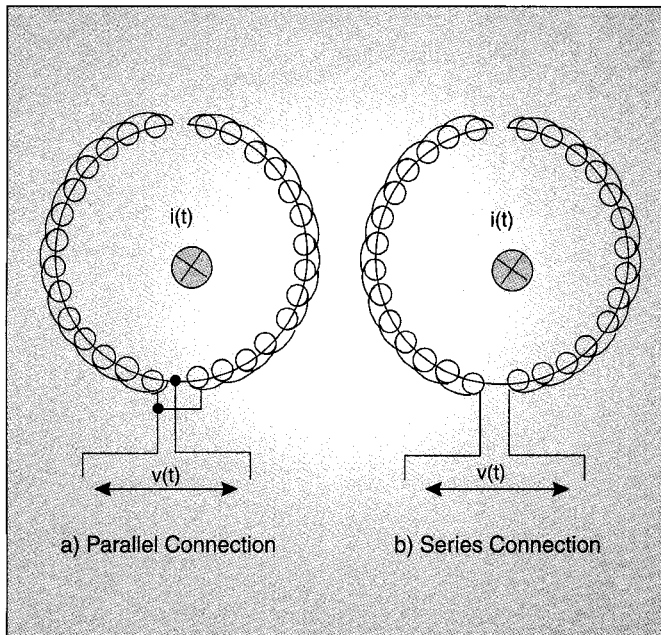


Figure 4. Two types of two-arm Rogowski coil designs

coils must be designed with two wire loops connected in the electrically opposite direction. This will cancel all electrical fields coming from outside the coil loop. This other loop can be formed by returning the wire through the center of the winding as shown in Figure 2 or by adding an additional winding wound in the opposite direction over the existing one.

To obtain a voltage proportional to the measured current, the coil output voltage must be integrated. Output voltage integration can be performed using an RC integrator or an operational amplifier as shown in Figure 3. The coil integrated voltage is proportional to the sum of the measured currents as given by Equation 4.

$$v(t) = \frac{M}{RC} \sum_j i_j(t) \quad (4)$$

The coil output voltage does not need to be integrated to measure sinusoidal currents. This is because the differentiation of sinusoidal currents results in sinusoidal waveforms shifted by 90 degrees, as shown by the following equations. This shift does not affect measurement results if only currents are measured. The coil output voltage is given by Equation 5.

$$\begin{aligned} i(t) &= I_m \sin(\omega t) \\ \frac{di}{dt} &= \omega I_m \cos(\omega t) \\ v(t) &= -\mu_0 n S \frac{di}{dt} \end{aligned}$$

$$V_{rms} = \mu_0 n S \omega I_m = \mu_0 n S \omega \sqrt{2} I_{rms}$$

(5)

Designs of Rogowski Coils

To obtain high precision Rogowski coils, the following design criteria must be applied:

- Constant winding density ($n = \text{constant}$)
- Constant coil cross-section ($S = \text{constant}$)
- Winding cross-section perpendicular to the middle line.

There are several different coil designs. Figure 2 shows a single-arm coil. Two-arm coils are shown in Figure 4.

Measuring error is increased if the primary conductor is located near the coils ends. This is because of winding discontinuity at the coil ends. To reduce measuring errors, coil ends can be shielded using standard blocks of transformer sheets as shown in Figures 5 and 6. This shielding can be removable to make coil placement easier.

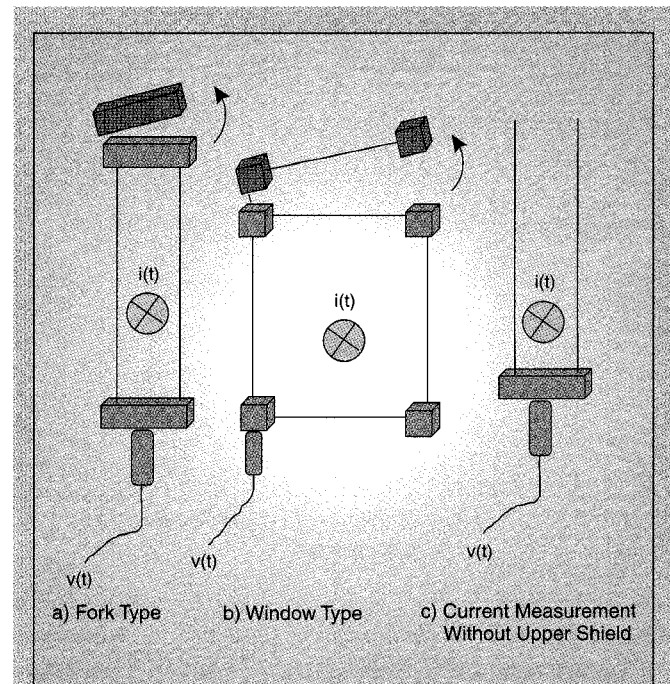


Figure 5. Rigid type Rogowski coils

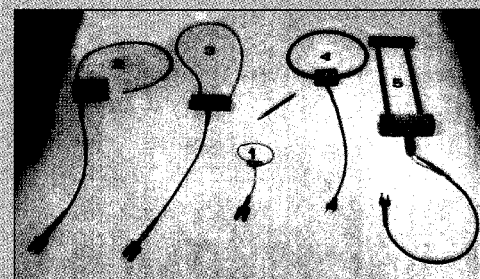


Figure 6. Various Rogowski coil designs

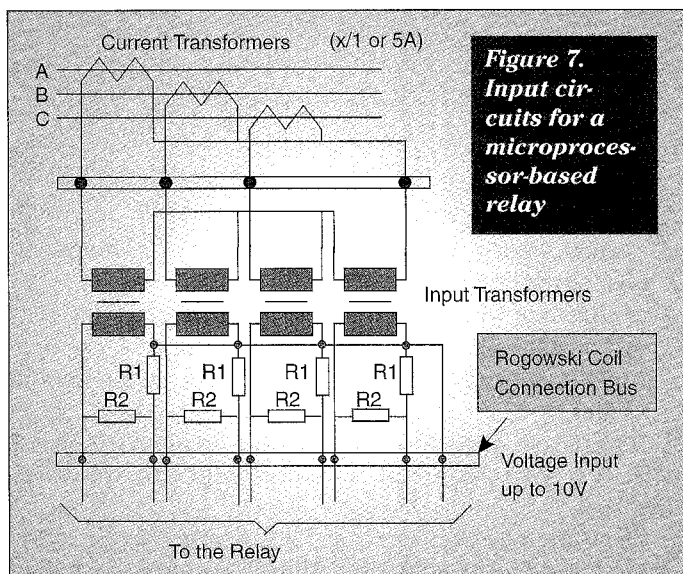


Figure 7.
Input circuits for a
microprocessor-based
relay

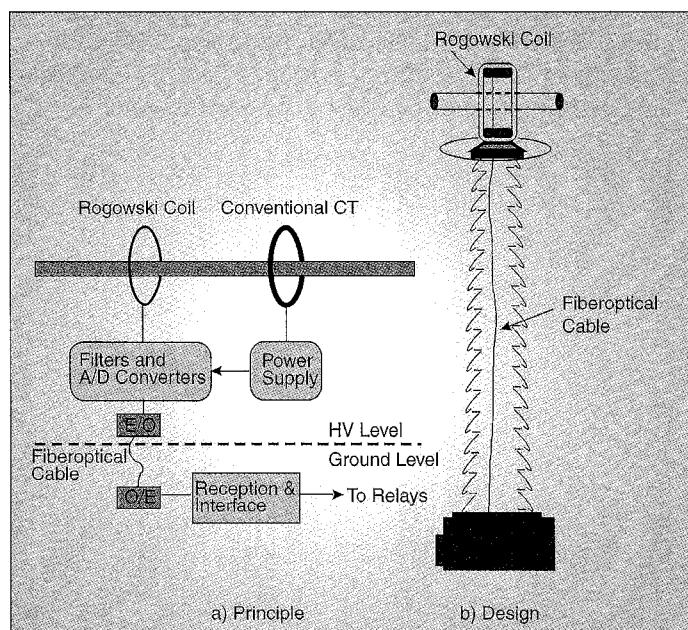


Figure 8. Application of Rogowski coils for current measurements in HV switchgear

High performance coils are designed by winding wire on straight rods. Cross-section area can be manufactured more uniformly with rods than with flexible cores. It is also easier to control the winding process on rods than on flexible cores. Fork-type and window-type coil designs are shown in Figures 5a and 5b. Fork-type coils without an upper shield can also be used to measure currents in conductors (Figure 5c) if there are no other nearby conductors carrying high currents. In this application, higher accuracy is obtained if the conductor is well inside the coil fork. Although flexible coils are less sensitive and less accurate than the rigid form coils, they are more compact and versatile and can provide acceptable performance. Coil sensitivity can be determined by measuring mutual inductance.

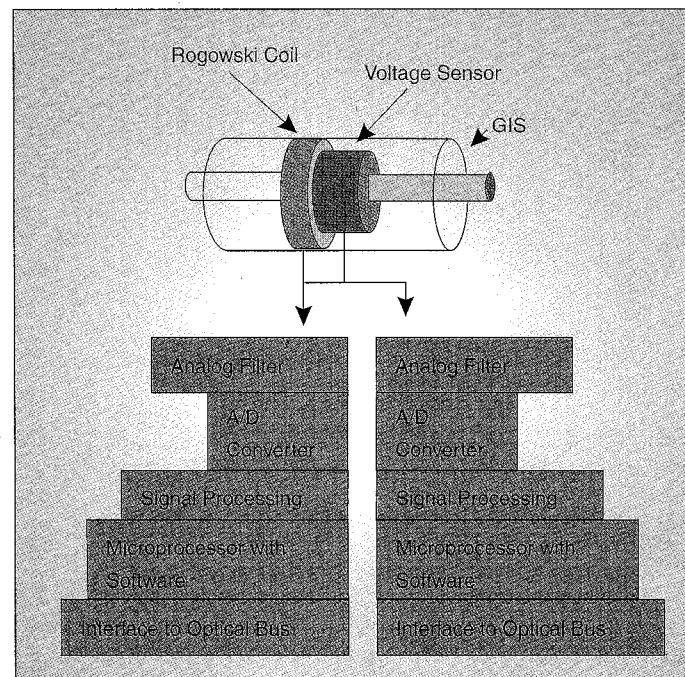


Figure 9. Application of Rogowski coils in GIS

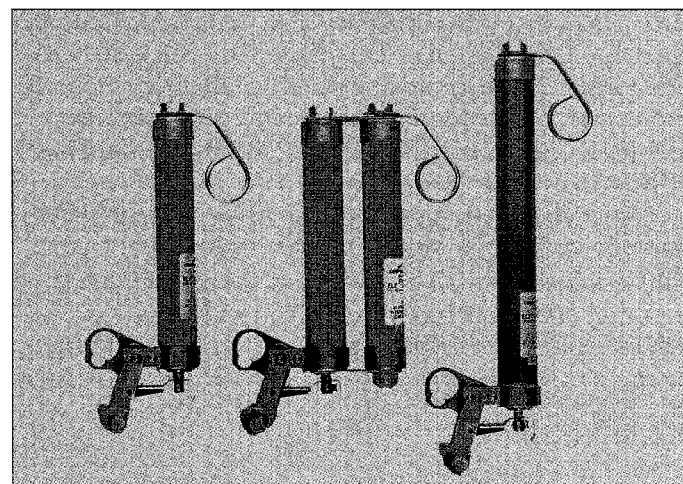


Figure 10. Single-barrel and multibarrel fuses

Figure 6 shows five different types of Rogowski coil designs compared in size to a pencil. They can be designed to be very small so they can be installed in locations where access is limited. They can also be designed to be very light in weight so that they can be suspended on the conductor being monitored.

Rogowski Coil Applications

Rogowski coils were introduced in 1912 to measure magnetic fields. At that time, they could not be used for current measurements since the coil output voltage and power were not sufficient to drive measuring equipment. With today's microprocessor-based protection and measurement equipment, Rogowski coils are more suitable for such applications.

Relay Protection

Microprocessor-based relays do not require high power current and voltage sensors. To accept signals from CT secondaries, microprocessor-based relays have scaling transformers at their inputs that transform standard secondary currents (1 A or 5 A) to required low voltage levels (Figure 7). Voltage levels at the input transformer secondary side which supplies microprocessor-based equipment is in the range of 5 V for 1 A input. Rogowski coils need to provide voltage at or below this level since the signal can be amplified during the integration of the coil voltage output or it can be done by the equipment itself. Rogowski coil output voltages can be connected directly to the relay voltage bus without the need for input transformers (Figure 7).

In the early 1980s, research into the use of different types of Rogowski coils for relay protection in medium voltage (MV) switchgear was conducted. Recent applications of Rogowski coils as an alternative to conventional CTs for MV switchgear have been reported, including one paper that compares designs of MV switchgear having conventional CTs and PTs with new MV switchgear designs which employ combinations of Rogowski coils and voltage sensors. The voltage sensors are resistive dividers. The application of Rogowski coils with digital processing equipment for protection and measurement in high voltage (HV) switchgear has also been investigated. The principle of current measurements presented in one study is shown in Figure 8.

New technologies for gas insulated substation (GIS) designs can also include Rogowski coils for current measurement. Microprocessor-based equipment will replace traditional equipment for protection, control, and measurement. Connections between equipment will be optical fiber cables. Rogowski coils are designed for use as current sensors. Voltage measurements are performed using the specially designed capacitor sensors shown in Figure 9.

Another application of Rogowski coils for protection systems is the measurement of low frequency slip currents on wound rotor induction motors. Rogowski coils are used for this application because they cannot saturate. Ground faults are detected by summing the three phase currents and looking for a nonzero result.

Other Current Measurement Applications

Rogowski coils have been used for a variety of current measurement applications. The following examples take advantage of the unique characteristics of Rogowski coils.

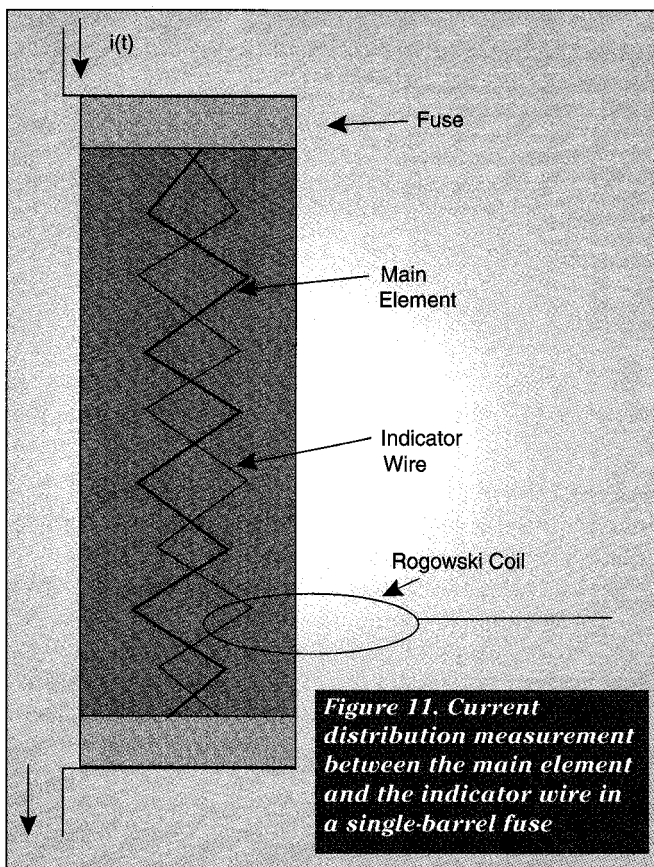
Measurement of current distribution in parallel fuses. Current-limiting fuses can use single-barrel or multibarrel designs. Multibarrel designs use two or three fuses connected in parallel to increase rated current (Figure 10). Rogowski coils were used to measure current

distribution between the main element and the indicator wire inside a single-barrel dropout fuse (Figure 11). For this application, an especially small Rogowski coil was designed to embrace the indicator wire through two holes that were made in the fuse tube. Test results were used to optimize fuse performance characteristics. The distribution of currents between fuses during multibarrel fuse operation is important since the fuses need to share energy absorption during faults. Figures 12a and 12b show the measurement method and current waveforms recorded during laboratory tests of a multibarrel fuse operation.

Current distribution in parallel bus bars or flexible braids. These are low impedance circuits. Rogowski coils do not affect current distributions. The setup of a measurement example is shown in Figure 13. Phase shifting between currents in the three parallel bars was caused by mutual reactance influences.

Current measurements in communication towers or power poles caused by lightning strokes. Rogowski coils can be designed to be long enough to embrace a communication tower and measure the total current through the tower. A 15 m long Rogowski coil was designed for such an application using a parallel connection as shown in Figure 4a to obtain accurate measurements at high frequencies.

Current measurements in arc furnaces. Currents in arc furnaces are fast changing and can have magnitudes



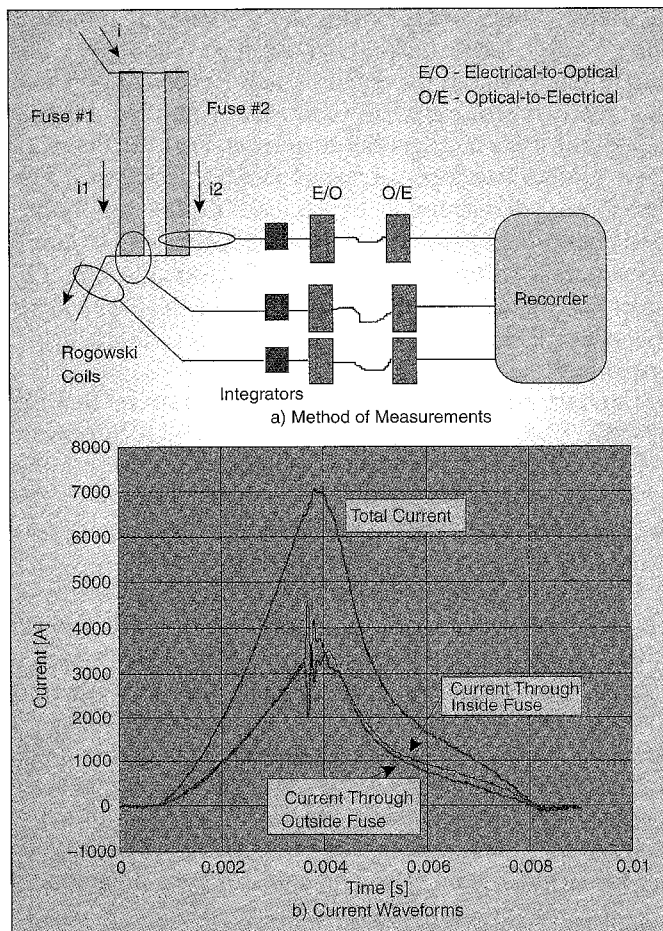


Figure 12. Current distribution measurement in multi-barrel fuses during testing

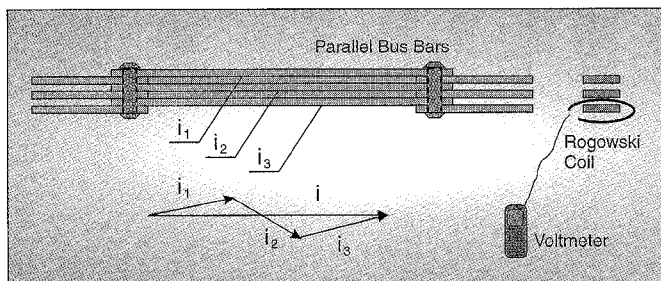


Figure 13. Current distribution measurement in parallel bus bars

between zero and 50 kA or 100 kA. Rogowski coils do not saturate and can also be designed to be long enough to embrace the large secondary leads.

Measurements of induced currents in the enclosures of GIS switchgears. High currents can be induced in GIS enclosures. Rogowski coils were used to measure the current distribution through GIS interconnections, through the grounding leads, and through the gas pipes.

Current measurements in generator main output connections. The enclosures that surround the conductors can be greater than 1 m in diameter. Rogowski coils can measure currents induced in these enclosures and their distribution.

Measurements of spurious currents in transformer pipework. Rogowski coils have been used to determine currents flowing in the cooling pipework of a large transformer which can caused heating at bolts and flanges.

Generator core framework current measurements. Due to their small dimensions, Rogowski coils can be installed in generator stators to measure the currents flowing around the stator support structure. This can provide data for research into stator hot spots and core faults to improve stator design.

Current measurements during sudden short-circuit testing of generators. The wide measuring range of Rogowski coils make them suitable for this application.

Current measurements during testing uninterruptable power supplies (UPS). Rogowski coils can be used to determine UPS recovery times after faults had been applied.

Advantages

The main advantages of Rogowski coils are: high measuring accuracy, wide measuring range, wide frequency range, ability to withstand unlimited short-circuit currents, design flexibility (e.g., very small to measure currents in the restricted areas where other techniques cannot be used; and flexible or rigid depending of application requirements), applicable for measurements of current distributions in circuits having very small impedance without affecting the circuits, galvanically isolated from the primary conductor, and are low in cost.

For Further Reading

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Biography

Ljubomir Kojovic received his Dipl. Ing. degree from the University of Sarajevo, Yugoslavia, MS degree from the University of Belgrade, and PhD degree from the University of Sarajevo, all in electrical engineering in 1972, 1977, and 1981, respectively. From 1972 to 1991 he was with the Energoinvest Company, Sarajevo, Yugoslavia. He was also an assistant professor at the University of Tuzla, Yugoslavia. In 1991 and 1992, he was a post-doctorate research associate at Texas A&M University. Since 1993, he has been a senior staff engineer for Cooper Power Systems at the Thomas A. Edison Technical Center. His name is included in the roster of experts for United Nations Development Organization (UNIDO). He is an IEEE Senior Member and a registered professional engineer in Wisconsin.